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BUNCH LENGTH REGULATION IN THE LHC DURING CONTROLLED EMITTANCE BLOW-UP

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Abstract

Controlled longitudinal emittance blow-up is indispensable for the operation of the Large Hadron Collider (LHC) to counteract single-bunch loss of Landau damping during the acceleration ramp. The blow-up is performed by injecting RF phase noise in a narrow frequency band into the beam phase loop, with bunch-length feedback regulating the noise amplitude. In 2024, the variation of the bunch length due to imperfect regulation caused unacceptable beam-induced heating of certain accelerator components. In this contribution, we present the results of extensive simulation scans that have been used to optimize the feedback parameters. We show how this optimization, along with a reduction of the feedback delay on the controls side, has been implemented in the LHC and significantly improved the bunch length evolution during acceleration. Finally, we discuss the results of a measurement scan performed during an operational period of five weeks to fine-tune the blow-up feedback settings.

INTRODUCTION

Controlled longitudinal emittance blow-up is essential in the LHC to ensure beam stability throughout the acceleration ramp. To maintain a constant stability margin, the longitudinal emittance ε must scale with beam energy according to $\varepsilon \propto \sqrt{E}$ [1]. It keeps a nearly constant bucket filling factor and relative synchrotron frequency spread. In practice, this is achieved using a feedback mechanism that adjusts the r.m.s. amplitude of the RF phase noise $\varphi_N(t)$ to keep a constant bunch length [2].

The applied noise spectrum, $S_\varphi(f)$, determines the frequency range in which diffusion occurs within the bunch, thereby influencing its final shape. In the LHC, a band-limited white noise spectrum is desired, targeting the core of the bunch by tracking the central synchrotron frequency, f_{s0}, along the ramp in the noise band of $(0.8571 - 1.05)f_{s0}(t)$. In practice, $\varphi_N(t)$ is injected via the beam phase loop (PL) as an additional phase shift in the phase correction between the 400 MHz bunch and cavity phase. However, as the phase loop damps oscillations around f_{s0}, the injected noise spectrum is pre-shaped to compensate the phase loop response [3].

The bunch length feedback is implemented using real-time bunch length measurements from the LHC Beam Quality Monitor (BQM). The algorithm compares the instantaneous measured bunch length τ_{meas}, average of all bunches in the ring, to the desired target length τ_{targ}, and scales the phase noise excitation amplitude with a dimensionless factor $x_n \in [0,1]$ as follows [4]:

$$\begin{aligned} &x_{n+1} = a x_n + g(\tau_{\text{targ}} - \tau_{\text{meas}})\,, \\ \text{if} \quad &x_{n+1} \le 0, \quad \text{then} \quad x_{n+1} = 0\,, \\ \text{if} \quad &x_{n+1} \ge 1, \quad \text{then} \quad x_{n+1} = 1\,, \end{aligned} \tag{1}$$

where n is the iteration. The recursion represents a low-pass filter (LPF), which suppresses high-frequency noise on the bunch length measurement. The 'memory factor' $a \in (0,1)$ is the weight of the previous x_n measurement and determines the time constant of the LPF. The gain g, typically $\mathcal{O}(1)$ ns^{-1}, weighs the bunch length difference.

In 2023 and 2024, non-conform RF finger modules have been leading to local heating in beam pipe bellows [5]. As a consequence, bunch length and intensity limitations have been imposed to ensure safe operation. For the bunch length, the controlled emittance blow-up was the bottleneck during the LHC cycle, with bunch length dips down to 1.0-1.1 ns, from the desired minimum of 1.2 ns. In this paper, we present the optimization of the blow-up, which led to smoother bunch length regulation and enabled us to maintain the bunch length above 1.2 ns.

SIMULATION MODEL

An extensive feedback parameter scan was executed in simulations, in order to determine the optimum feedback parameters a and g from Eq. (1), which would produce a smooth bunch length evolution. Each ramp in the LHC takes about 2h and the experimental optimization of the feedback parameters would have been impossible. The longitudinal tracking suite BLonD [6] has been extended to more accurately mimic the behaviour of the noise feedback system, specifically for the LHC implementation. The use of GPUs was imperative to execute such a scan, since each simulation of the entire LHC ramp (1.4×10^7 revolutions or ~25 min) is very computationally heavy.

Simulation Conditions

Due to the high computational cost of the feedback optimization, one simulation per (a, g)-parameter pair was performed, using fixed initial conditions and a single bunch. In reality, the bunch distribution and multi-bunch effects can have a second-order effect on the bunch length evolution.

The single bunches were modelled using 10^6 macroparticles representing an intensity of 1.6×10^{11} p/b (protons per bunch). Intensity effects were taken into account, applying the latest LHC impedance model up to 6.25 GHz. The acceleration ramp was simulated using the operational energy and voltage programs from 450 GeV to 6.8 TeV and

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from 5 MV to 12 MV, respectively. The initial bunch length at flat bottom was 1.36 ns, which corresponds to the average of operational values in 2024. The target bunch length was set to 1.3 ns throughout the ramp.

The initial bunch shape followed a binomial distribution and the bunch length was derived from the simulated bunch profiles with the same algorithm as applied by the BQM. The full-width-half-maximum (FWHM) is scaled to $\sqrt{2/ln2}$, sampling at 8 GS/s. To further mimic the real execution in the LHC, the bunch length evaluation took place every 1.1 s and the noise scaling was done with buffers of 1 s. The new x_{n+1} was calculated according to Eq. (1) from the bunch length measurement occurring during the penultimate noise buffer. The bunch length measurement and the noise scaling had an initial fixed latency of ~1.02 s with respect to each other. This was sampled randomly with a fixed seed from the delay that can vary between 0 s and 1.1 s.

The phase noise array was pre-generated with the operational noise spectrum in the range $(0.8571 - 1.001) f_{s0}(t)$, and then injected through the simulated beam phase loop. More details on the exact simulation procedure can be found in [7].

GPU Acceleration

The BLonD GPU implementation gave a significant speed-up for the simulation scan [6]. The usual CPU time for simulating a single ramp is approximately one week. On the other hand, the GPU runtime is somewhere between 10 h to 18 h, depending on the GPU model, resulting in a total speed-up of factor about 12. A total of 177 different (a, g) parameter pairs were simulated, summing up to a total run time of ~2478 h on GPU.

SIMULATION RESULTS

Different pair combinations of the feedback parameters were used in the simulations: memory factor a, range (0.65 – 0.975), and gain g, range (0.05 – 0.6) ns^{-1}.

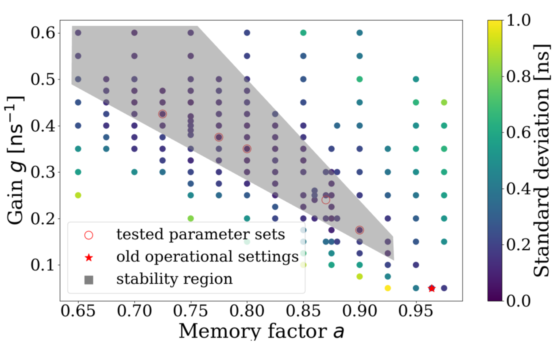

Figure 1: Stability region of the feedback parameters based on standard deviation during the ramp

The effectiveness of the bunch length regulation was evaluated using the standard deviation of the bunch length measurement from the effective target bunch length of 1.27 ns as a metric, see Fig. 1. The effective target bunch length (1.27 ns) is slightly lower than the target bunch length used for regulation (1.3 ns), due to the finite gain of the negative feedback loop. The standard deviation was measured from the moment the blow-up mechanism is activated ($x_n > 0$), to exclude the initial adiabatic shrinkage of the bunch. For a clearer illustration of the results, simulations with a standard deviation larger than 5×10^{-2} ns have been excluded from the plot.

This analysis revealed a region of optimum feedback parameters for a stable bunch length regulation (grey region in Fig. 1). This area has been selected taking into account only the simulation results that have a standard deviation below 2×10^{-2} ns.

OPERATIONAL OPTIMIZATION

In the past, the non-ideal bunch length regulation led to regular undershoots of the bunch length to 1.0-1.1 ns in the beginning of the ramp (at around 200 s after the start of acceleration), followed by large overshoots of up to 1.6 ns as of the second third of the ramp (from 500 s onwards).

The operational settings in the LHC at the start of beam commissioning in April 2024 were as follows. The software implementation for noise injection had a buffer length of 2 s and a reaction time to a new measurement of two buffers, with respect to the time of the bunch length measurement [8]. The LHC BQM acquired a profile measurement every 1.1 s. This means that the expected delay between the bunch length measurement and the new buffer containing noise with corrected amplitude was between 4 s and 6 s, corresponding to 4.5×10^4 and 6.7×10^4 machine turns. The feedback parameters were set to $a = 0.964$ and $g = 0.05\ \text{ns}^{-1}$.

While simulation studies were starting, a quick mitigation of the bunch length undershoots was obtained by increasing the bunch length target at the start of the ramp. For the first 200 s it was set to 1.5 ns, followed by 1.4 ns until 500 s, then 1.35 ns until 800 s, and finally 1.33 ns until the end of the ramp. With this target function, the bunch length was kept above 1.2 ns.

First Tests in the LHC

The first optimizations of the bunch length regulation took place in June 2024, during so-called Machine Development (MD) sessions. Due to the long ramping time of the superconducing magnets, only two full ramps were possible in the allocated machine time of 4h. To test the blow-up mechanism on multiple beam configurations simultaneously, we decided to use bunch trains (12 b + 3×36 b) in beam 1 and a single bunch in beam 2.

In the first ramp, fill 9727, a new software implementation was tested, to decrease the response time of the noise injection mechanism. The buffer size was reduced to 1 s and the response time to one buffer. The expected delay between the BQM measurement and the action on the noise buffer is now between 1 s and 2 s, corresponding to 1.1×10^4 and 2.2×10^4 turns. This change already improved the bunch length regulation visibly, see Fig. 2, fill 9727.

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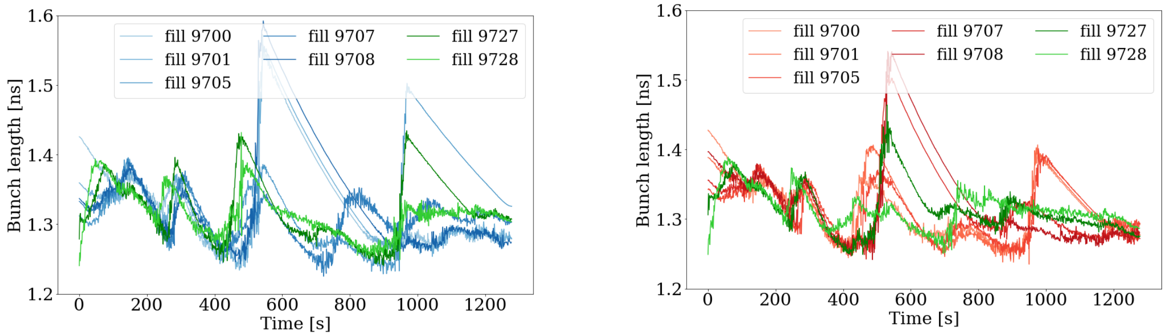

Figure 2: Bunch length evolution during the LHC ramp before (fills: 9700 – 9708) and during the MDs (fills: 9727, 9728). Beam 1 (left) and beam 2 (right).

Table 1: Feedback Settings During the LHC Parameter Scan

Settings	WK 35	WK 36	WK 37	WK 38	WK 39
$a[1]$	0.9	0.87	0.8	0.775	0.725
g [ns^{-1}]	0.175	0.24	0.35	0.375	0.425

In the second ramp, fill 9728, a new pair of feedback parameters was tested along with the new software implementation. The parameters $a = 0.87$ and $g = 0.2$ ns^{-1} were chosen based on preliminary simulation results. With this change, the bunch length can be regulated faster, as is visible especially at the start of the ramp in Fig. 2, fill 9728. Both changes resulted in a significant improvement of the bunch length control and were retained for LHC operation right away.

Target Bunch Length Changes

Thanks to the improved settings from the MDs, the stepwise target bunch length had become unnecessary and even seemed to deregulate the bunch length evolution. In operational ramps (different beam configurations), the target was lowered by 0.05 ns per ramp, in a total of four steps (four days), until it became flat at 1.35 ns.

The effect of the target function on the bunch length evolution is presented in Fig. 3. One can observe that the closer we get to the flat target, the more natural the adiabatic shrinkage is at the beginning of the ramp. In all cases the bunch length never drops below 1.2 ns, fulfilling the requirement to avoid RF heating.

Operational Scan

Once the extensive feedback parameter simulation scan was completed, an operational scan with beam was conducted in the predicted stable region shown in Fig. 1. The goal of this scan was to validate the stable region and attempt to further fine-adjust the feedback settings.

Each parameter set was kept for a week and tested on different beam configurations, to investigate also the fill-by-fill variations. The parameter pairs along with the corresponding week number in 2024 are summarized in Table 1.

Each set of parameters gave similar bunch length regulation, as expected, confirming the predicted region. Finally, it

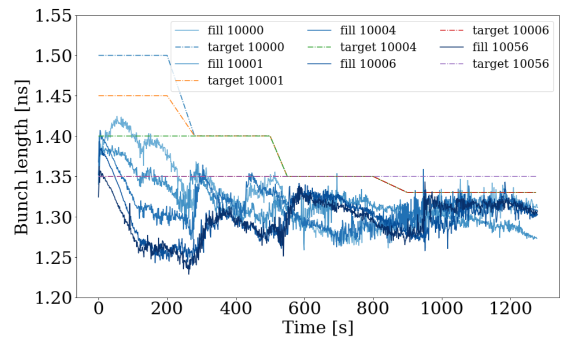

Figure 3: Bunch length evolution during the change of the target bunch length function (beam 1).

was decided to keep the pair with the higher gain ($a = 0.725$ and $g = 0.425$ ns^{-1}), as it is preferred that the feedback has a faster reaction time according to more recent bunch length measurements.

CONCLUSIONS

The bunch length regulation in the LHC was suffering from large excursions with respect to its target value during the acceleration ramp. Thanks to simulations with GPU resources, it was possible to scan a large range of the phase noise feedback parameter pairs and map the stable region. In addition, with improvements in the LHC software implementation of the phase noise injection, the latency between bunch length measurement and noise amplitude regulation was reduced.

After changing the feedback parameters and adopting the new software implementation in the machine, the bunch length control improved significantly. A flat target bunch length function was also restored. Finally, five pairs of feedback parameters were tested in operation, confirming similar regulation quality as predicted. The pair with the highest gain ($a = 0.725$ and $g = 0.425$ ns^{-1}) was kept for operation. This way, local heating on the machine induced by the RF finger modules is no longer an issue for the present operational intensity of 1.6×10^{11} p/b.

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